

Shiv Vyas

Software Engineer | Full Stack, iOS, React Native, Applied AI | TypeScript, Swift, Python, SQL

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Summary

Software engineer with an **M.S. in Computer Science** who has **designed, built, tested, shipped, and maintained** products across the full stack: web (**TypeScript, React, Next.js**), back end (**Node.js, Python, PostgreSQL**), and mobile (**Swift, React Native**) for **20,000+ users**. Founding engineer twice over, comfortable owning a product from an empty repository to an App Store release, and increasingly building with **LLMs in the loop**, on device where possible and in the cloud where necessary.

Experience

Software Engineer, Contextual Intelligence (consumer social platform, formerly Luna Social) December 2024 – Present

- **Design, develop, test, deploy, and maintain** the product across web (**React, Next.js, TypeScript**) and native **iOS** (Swift) for **20,000+ users**; largest of 15 contributors on the mobile codebase (**474 of 2,117 commits**) and author of **141 of 145 API route handlers** (payments, signup, SMS verification, onboarding, events)
- Built a **distributed serverless back end** for correctness under concurrency: atomic **database transactions** for recommendation and connection writes, Stripe **webhooks idempotent under retries** keyed by event id, and a scheduled health job that probes every dependency and **alerts on degradation**
- Applied **data structures** to performance: a **500 MB LRU disk cache** keyed by **SHA-256** with coalesced downloads so ten requests for one asset make one network call, plus a bounded preloader, so the video feed stays responsive on cellular
- **Improved engineering velocity and quality**: **1,218 unit tests** across 109 Vitest suites, XCTest and XCUITest gates in **CI/CD**, **secret scanning** in GitHub Actions, code review on every pull request, and **accessible UI** (ARIA on web, VoiceOver and Dynamic Type on iOS)

Founding Software Engineer, Futeur AI (fintech: business credit, payments, funding) August 2024 – May 2026

- **Designed, built, and released** FuteurCredx solo to the **App Store and Google Play** (React Native, TypeScript), growing downloads **84% within 30 days** of launch; managed priorities and deadlines from 1.0 through 1.2.2
- Integrated three credit-bureau feeds, **Plaid**, and **Stripe**, with per-source cache windows and **polling only while an application is in flight** to cut redundant network traffic; offline persistence and **124 Jest tests across 16 suites**
- Delivered white-label apps for **Chase, Wells Fargo, Citibank, and Santander** from one shared codebase, so each brand was a theme rather than a fork, coordinating each release with the partner's engineering team

Lab Coordinator, Seidenberg School of CSIS, Pace University April 2024 – September 2024

- Replaced a manual request process with a **self-serve GPU compute platform** for **400+ students**, then administered the shared servers, triaging compute, environment, and system health issues across two campuses

Software Developer Intern, Ordex Technology February 2023 – July 2023

- Cut rendering times **60%** on the Citizex mobile app by profiling and refactoring the render path, and optimized authentication flows, lifting sign-up **5%** and engagement **20%**

Projects

Sanchara: AI Astrology Platform, Web and iOS (Next.js, TypeScript, PostgreSQL, Swift, Claude API) 2026

- Built one **API shared by a web app and a native iOS app**: a pure computation engine (Swiss Ephemeris via **WebAssembly**) derives charts, time periods, and transits, and an LLM only interprets computed values, streamed to the client; **209 unit tests** across 25 Vitest files
- Applied **natural language processing** to extract life events from users' own chat history with the quote each came from; scheduled jobs send push notifications on each user's local clock; **row-level security** in PostgreSQL

CinemaHUD: Networked Camera Monitor for Mac, iPhone, and iPad (Swift, Python, Metal) 2026

- **Networking**: a Mac server advertised over **Bonjour** streams live video, server-sent state, and remote-control commands to iPhone and iPad clients; drives the camera over **USB (PTP)** and Wi-Fi with automatic reconnect and newest-frame-wins buffering
- **Algorithms and parallelism**: **ring buffer** audio capture and **normalized cross-correlation** to sync separately recorded audio to video; a GPU pipeline that cut CPU load from **72% to about 14%**; **166 tests** and a Python camera simulator

Life OS: iOS App over a TypeScript and PostgreSQL Back End (Swift, TypeScript, SQL) 2026

- **13 PostgreSQL migrations**, 13 serverless functions, lease-based **concurrency control**, and crash-safe cursor sync; on-device **machine learning** (Apple Foundation Models) with a cloud fallback; **1,298 tests across 154 suites**, delivered spec-first with Claude Code

Technical Skills

Languages	TypeScript, JavaScript, Python, Swift, SQL , Kotlin, HTML, CSS, Bash
Web & Back End	React, Next.js, Node.js , FastAPI, PostgreSQL , Firebase, Supabase, REST, server-sent events, webhooks, OAuth 2.0, AWS (S3, CloudFront), Vercel
Mobile	iOS (SwiftUI, UIKit, Combine, Swift Concurrency), React Native (Expo), Android (Kotlin, Jetpack Compose)
Systems	Unix/Linux shell, networking (Bonjour, TCP, HTTP), concurrency and parallelism , database transactions, caching, GPU (Metal), security (row-level security, secret scanning)
AI & Tools	Claude Code, Cursor , spec-driven development, LLM APIs (Claude, OpenAI, Gemini), streaming, structured output, Git, GitHub Actions, Jest, Vitest, XCTest, Playwright

Education

May 2025	Master of Science in Computer Science , Pace University, New York (GPA: 3.74/4.0) Coursework: Data Structures and Algorithms, Parallel Programming, Database Management Systems, Human-Computer Interaction
June 2023	Bachelor of Engineering in Computer Engineering , Gujarat Technological University (GPA: 3.70/4.0)